# Ardhya Novri Ramadhan

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Bachelor of Engineering in Electrical Engineering from Diponegoro University, with experience in electronics, airport electrical systems, student organizations and community projects. Develops embedded monitoring systems and AI-assisted software.

## EDUCATION

**Diponegoro University** Aug 2022 - Sep 2026
*Bachelor of Engineering in Electrical Engineering*
GPA: 3.43/4.00. Graduated on 29 September 2026.

**SMA Negeri 5 Bekasi** Jul 2019 - May 2022
*Natural Sciences*

## PROFESSIONAL AND COMMUNITY EXPERIENCE

**Diponegoro University** Jan 2026 - Feb 2026
*Community Service Participant (KKN) | Mutihan, Klaten*

• Helped assemble and install a solar-powered Smart Owl House, including sensors, battery and controller. Prepared operating and maintenance guidance.
• Assisted village administration, Posyandu activities, BUMDes bookkeeping and community cleanups. Joined mosque and cemetery painting activities.
• Supported school scouting activities and earthquake preparedness education. Helped produce and hand over the village profile video.

**PT Angkasa Pura Indonesia** Jul 2025 - Aug 2025
*Internship Trainee*

• Rotated for five weeks in UPS/converter systems and three weeks in electrical protection at Soekarno-Hatta International Airport.
• Visited substations and focused on learning protection relays. Received SCADA instruction and observed PLC installations.

**Diponegoro University** 2024 - 2025
*Basic Electronics Practicum Assistant*
Served for two consecutive semesters in the 2024/2025 academic year, one academic year in total. Assisted students with laboratory activities and basic electronic components.

**Redesma Technologies** Jan 2025 - Feb 2025
*Electronics Technician*
Worked on IoT temperature monitoring and DC motor control. Co-authored an SOP for comparing Blynk and Thinger.io using ESP32 and a DS18B20 sensor.

## SKILLS

**Electrical and embedded:** Electronics, sensor integration, C, STM32 HAL, STM32CubeIDE, ADC, DMA, timers and UART. Field learning in UPS/converters, protection relays and SCADA.

**Software and media:** Python, Tkinter, React, FastAPI, Git, automated tests, digital filtering, RMS/MAV extraction, CSV/JSON processing and Canva.

## ORGANIZATIONS AND EVENT COMMITTEES

**HME, Diponegoro University** 2025
*Secretary-Treasurer, Kantor Media Informasi (KMI)*
Served for one year in 2025, following the Creative Design Staff role in 2024.

**Society of Electronics Engineering Division (SEED)** 2025
*Head of Media Division*
Served as Head of Media for the 2025 student bureau at Diponegoro University.

**Google Student Ambassador Program** Class of 2025
*Google Student Ambassador*
Completed the student program and received recognition for championing Google AI within the campus community.

**Society of Petroleum Engineers (SPE)** Sep 2024 - Dec 2025
*Student Member*
Joined professional learning activities, including technical writing and geothermal drilling sessions.

**ENIGMA Vol. 3, SEED / HME UNDIP** Oct 2024
*Workshop Instructor*
Taught basic electronics workshop activities involving circuits, wiring and PCB work.

**Indonesia International Sustainability Forum (ISF)** Sep 2024
*Organizational Delegate*
Participated in the sustainability forum as an organizational delegate.

**Himpunan Mahasiswa Elektro, Diponegoro University** 2024
*Creative Design Staff, Kantor Media Informasi (KMI)*
Supported student organization media and design work. Assigned as a co-responsible staff member for the HME Rewind documentation program.

**LKMM-PD, HME UNDIP** 2023
*Event Committee Member*
Served on the committee for introductory student management and leadership training.

**PMB Teknik Elektro, Diponegoro University** 2023
*Logistics / Equipment Committee Member*
Prepared equipment and facilitated the needs of the new-student welcoming activities.

**Decarbonizing Indonesia** Nov 2022 - Jan 2023
*Campaign Contributor*
Created social media campaign and design materials about decarbonization.

**PKKMB and Pendikar, Diponegoro University** Aug 2022
*Participant*
Completed the 40-hour new-student orientation and character education program, 15-20 August 2022.

**Information and Communication Technology (ICT), SMA Negeri 5 Bekasi** Jan 2021 - Dec 2021
*Vice Secretary*
Prepared work-program and accountability documents, and maintained attendance and member assessment records.

**ICT Leadership Regeneration, SMA Negeri 5 Bekasi** Nov 2021
*Committee Secretary and Session Instructor*
Prepared event administration and delivered the secretarial training session for incoming ICT committee members.

**CINOTIC, SMA Negeri 5 Bekasi** 2021
*Event Secretary*
Supported proposals, reporting and event administration for the Science Techno & Journalistic Competition.

## PROJECTS AND TECHNICAL PROPOSALS

**RepairGate** Oct 2026

Developed a code-repair verifier with protected tests, independent holdout checks and exported patch decisions. Includes an optional local Ollama adapter and a recorded-candidate demonstration.

**ApplyProof** Oct 2026

Developed an application workspace with a reviewed fact ledger, explicit skill comparison, office/worksite distinction and traceable application drafts. Optional AI selects fact IDs for verbatim drafting.

**DataRescue** Oct 2026

Built a local CSV/JSON validation workbench and CLI with rule-based normalization, quarantine of rejected rows and audit-report exports, covered by automated tests.

**uart-fixture-kit** Oct 2026

Developed reusable synthetic UART fixtures and virtual-time replay to reproduce fragmented, malformed and concatenated byte streams without serial hardware.

**Personal Portfolio / Anak Kos Survival** 2026

Developed an engineering portfolio with project evidence and a simulated EMG preview. Earlier work includes a seven-day browser game and an interactive portfolio world, retained in the project archive.

**Nusantara SolarWing** 2026

Prepared a model-based concept package using historical solar data, wind sensitivity and battery reserve planning for maritime patrol scenarios.

**EMG Acquisition and Monitoring System** Thesis, 2026

Academic thesis, Diponegoro University.

- Implemented 12-bit ADC acquisition with DMA on STM32F411CEU6 and SEN0240 at a nominal 4 kHz sampling rate.
- Applied 20 Hz high-pass, 500 Hz low-pass and 50 Hz notch filtering, with RMS/MAV extraction over 400 samples.
- Built the Python/Tkinter interface for UART telemetry at 115200 baud, live plots, status monitoring and CSV export, with coding-agent assistance. Teammates handled the stimulator and fuzzy control.

**PLTA Jatigede Case Study** 2025

Contributed the introduction and task allocation to a group presentation covering NPV, benefit-cost ratio and payback period.

**Redesma Technologies / Timor-Leste Concept** 2025

Authored a proposal for rural technology options, including solar-powered irrigation, community warning systems and digital monitoring.

**EnvironmentID: IoT-powered Biofiltration** 2024

Co-authored an abstract proposing smart sensors and oyster ecosystems for coastal water-quality monitoring and biofiltration.

**GOVTECH / Gov-AI Hackathon Proposal** 2024

Prepared a proposal for integrated digital public services using APIs, responsive interfaces and AI-assisted information services.

## COURSES, WORKSHOPS AND CERTIFICATIONS

**Sep 2026:** Introduction to Internet of Things, Cisco Networking Academy. Completed 27 September 2026.

**Sep 2026:** AI Fundamentals, Google via Coursera. Completed 26 September 2026. Evidence is a course-completion screenshot, not an issued certificate.

**Sep 2025:** SPENTRODUCE: Exploring the Downstream Oil and Gas Industry, SPE UNDIP Student Chapter. Participant.

**Jan 2025:** AI-Powered Shopping Ads Certification, Google Skillshop. Historical credential, expired January 2026.

**Oct 2024:** Workover and Well Service Talks, IATMI Student Chapter ITB. Participant.

**Sep 2024:** Automation Clinic: Motor Protection and Control, Engineering Academy. Participant.

**Sep 2024:** How to Write a Good Technical Paper, SPE. 1.5-hour continuing education session.

**Sep 2024:** Geothermal Drilling in Volcanic and Metamorphic Basins, SPE. 1.5-hour continuing education session.

**Sep 2024:** Strategies in Geothermal Exploration: Maximizing Indonesia's Potential, SPE UNDIP Student Chapter. Guest lecture participant.

**Sep 2024:** Ethics of Artificial Intelligence: Navigating Responsible Technology for the Future, SkilHub Community by Skilvul / Nocturnailed. Webinar participant.

**May 2024:** Fundamentals of Digital Marketing, Google Digital Academy / Skillshop. Completed 27 May 2024.

**May 2024:** Introduction to Electronics, The Open University OpenLearn. Completed a 10-hour course on 21 May 2024.

**May 2024:** IT: Device to Device Communication, The Open University OpenLearn. Completed a 20-hour course on 19 May 2024.

**Mar 2024 - May 2024:** College Copilot Bootcamp (CABOOT). Completed six sessions, 20 March - 4 May 2024.

**Nov 2020 - Feb 2021:** KSN-K Geography Preparation. Participant in the 8 November 2020 - 7 February 2021 training program.

**Date not recorded:** ML/AI Fundamentals: How Machine Learning Is Shaping the Oil & Gas Industry, SPE Java Section.

## ADDITIONAL RECOGNITION

**May 2024:** 100 Design Achievement Badge, Canva.

**2019:** Try Out Akbar Winner, Penerbit Erlangga. Score: 100, SMP/MTs category, Jakarta. SMP Angkasa Halim Perdanakusuma.